Mahesh Chitakoti

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Summary

Backend & AI Engineer with 3+ years of software engineering experience building production RAG systems, LLM workflows, and backend services with **Python (FastAPI), TypeScript (Node.js), PostgreSQL, and Redis**. Built LangGraph workflows and Qdrant hybrid search for language learning and reduced recurring LLM inference costs by **35%**.

Technical Skills

Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, Express.js, AsyncIO, REST APIs, WebSockets

AI & Retrieval Systems: RAG Pipelines, Agentic Workflows (LangGraph), Qdrant, pgvector, Hybrid Search (Dense + BM25), Semantic Caching, Pydantic Structured Outputs, Gemini APIs, OpenAI APIs

Databases & Cloud Infrastructure: PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD

Experience

Independent Software Developer – Contract & Client Projects — Apr 2026 – Present

Remote

- Engineered an automated document ingestion and data-extraction backend using **FastAPI and Redis Queues**, processing multi-page PDF invoices and client webhooks with structured schema validation.
- Built a domain-specific RAG service using **PostgreSQL (pgvector HNSW indexing)** and semantic caching, cutting average vector retrieval latency from **280 ms to under 185 ms**.
- Containerized and deployed microservice workloads using **Docker on AWS EC2**, provisioning object storage in **S3** and establishing automated linting and test runs via GitHub Actions.

AI & Full Stack Developer – SahioNext Technology Pvt. Ltd. — Feb 2025 – Apr 2026

Remote

- Built Tulkka's lesson evaluation engine to generate structured feedback from lesson transcripts. Used **FastAPI and Gemini APIs**, with **LangGraph** for orchestration and **Pydantic** and fallback schemas to validate JSON output.
- Reduced recurring LLM inference costs by **35%** with exact-match caching and **Redis-backed semantic caching**. Mitigated upstream HTTP 429 errors with token-bucket rate limiting.
- Built hybrid search in **Qdrant** over **10,000+ lesson transcript chunks**, combining dense embeddings and BM25 to retrieve relevant vocabulary with p95 latency under **300 ms**.
- Developed full-duplex **WebSocket endpoints** for real-time speech evaluation, using client-side buffering and reconnection handling to recover interrupted sessions.
- Built internal dashboards in **TypeScript and Node.js** to curate datasets and combine reviewer rubric scores with automated model feedback.

Full Stack Developer – The DevOps Team — Oct 2023 – Jan 2025

Bangalore, India

- Profiled bottleneck queries and optimized **PostgreSQL indexing** alongside **Redis caching**, reducing median API response times from **800ms to 320ms (60% decrease)**.
- Redesigned candidate search queries using multi-column compound indexes, cutting database query latency and improving recruiter role-matching discovery rates by **22%**.
- Built responsive onboarding and profile workflows in **Next.js**, running A/B experiments on multi-step forms to lift user registration conversion by **18%**.

Projects

WorkBridge – AI Collaboration Platform — *Sep 2026 – Present*

Python, FastAPI, Node.js, PostgreSQL, pgvector, Redis, Next.js | Live Project

- Architected a decoupled microservices platform pairing a **Node.js API gateway** for session auth and RBAC with an asynchronous **FastAPI service** handling vector calculations and similarity search.
- Built candidate-to-project recommendation matching with **pgvector HNSW indexing**, executing sub-100ms cosine similarity retrieval against structured skill embeddings.
- Measured p95 search latency under **150 ms** with **k6** and 50 virtual users after tuning PostgreSQL connection pooling and Redis session caching.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering — *Aug 2023*